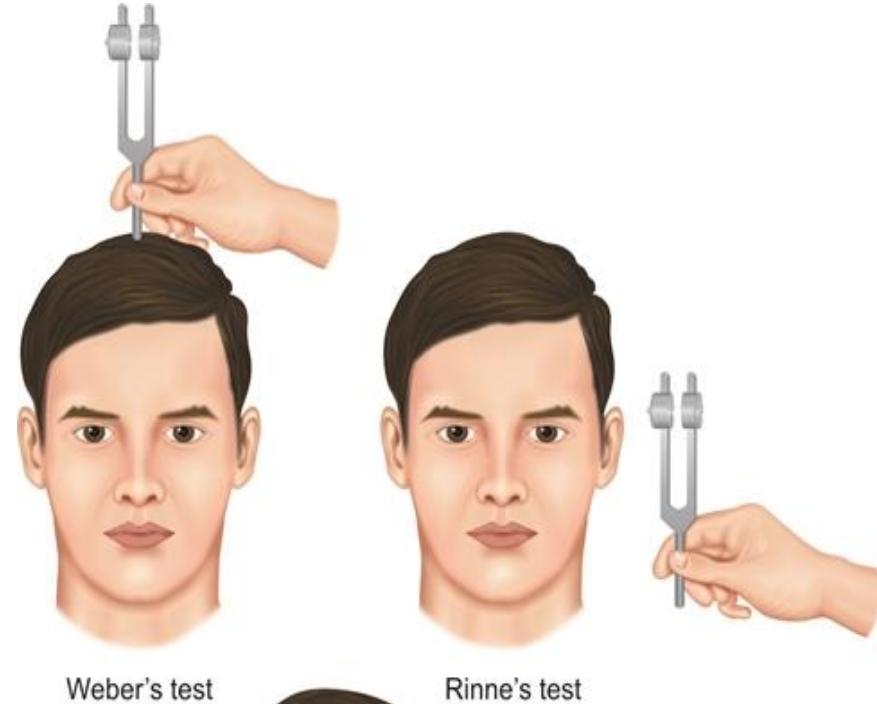


HEARING TESTS

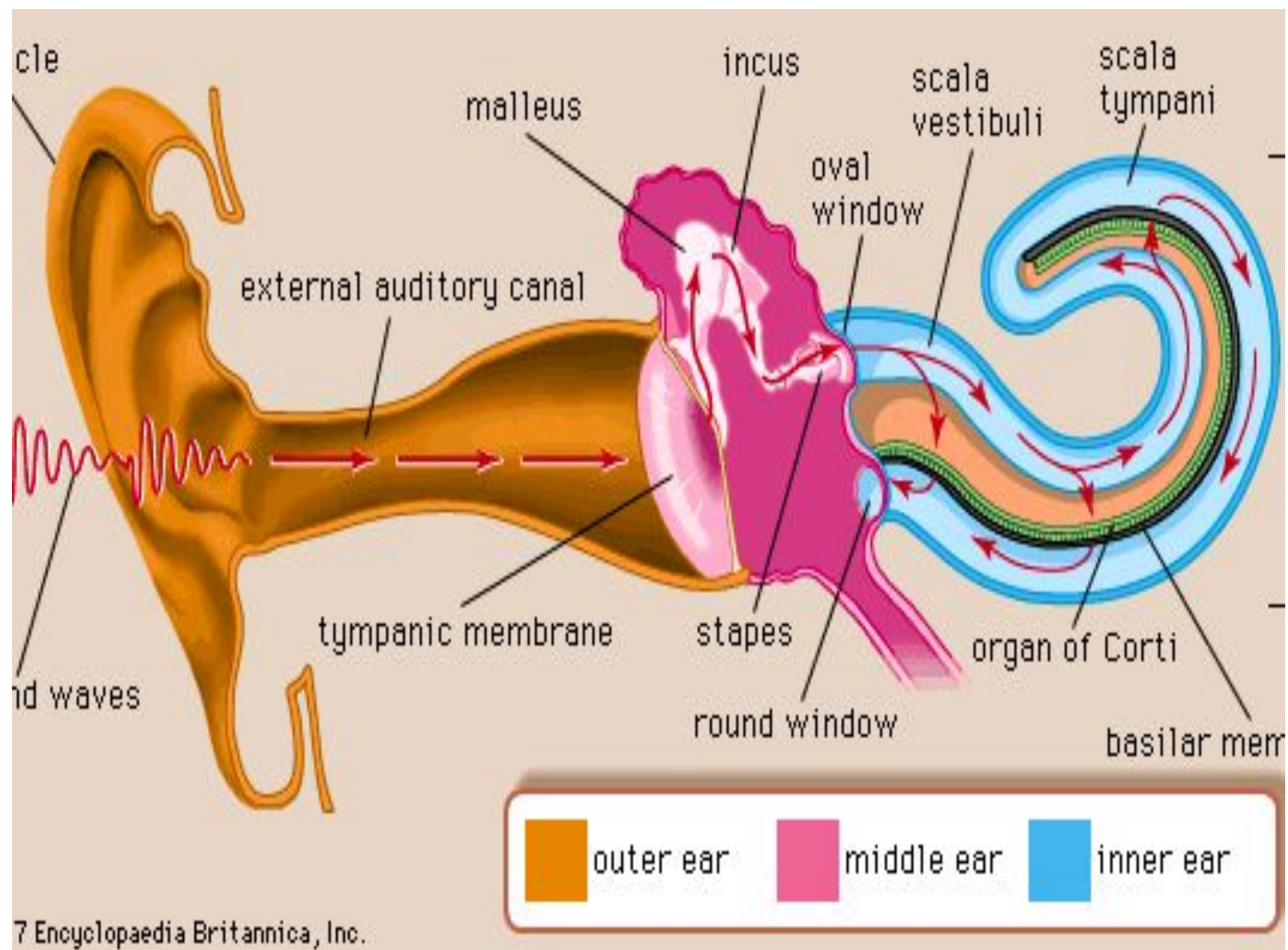


DR ABDUL SAMAD KHAN
SENIOR Lecturer
DIKIOS-DUHS

LEARNING OBJECTIVES

By the end of the session, the students should be able to:

- Describe the normal mechanism of hearing
- Differentiate between the types of deafness
- Perform the hearing tests
- Interpret the results



How We Hear

1.

Sound waves enter your outer ear and travel through the ear canal to your eardrum.

2.

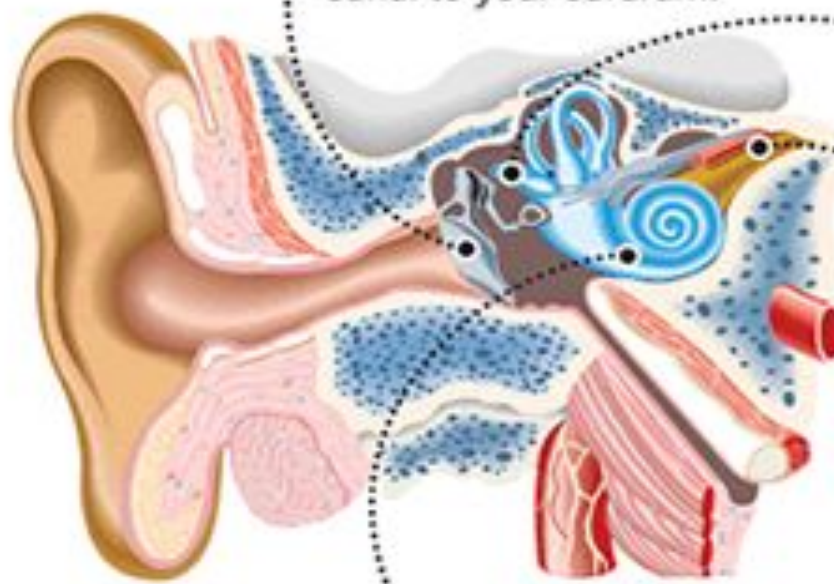
Your eardrum vibrates with the incoming sound and sends the vibrations to three tiny bones in your middle ear.

3.

The bones in your middle ear amplify the sound vibrations and send them to your inner ear, or cochlea. The sound vibrations activate tiny hair cells in the inner ear, which in turn release neurochemical messengers.

4.

Your auditory nerve carries this electrical signal to the brain, which translates it into a sound you can understand.



Mechanism of Hearing

The mechanism of hearing involves the following steps:

- The sound waves pass through the auditory canal and reach the eardrum.
- The vibrations produced pass through the tympanic membrane to the tympanic cavity.
- The ear ossicles in the tympanic cavity receive the vibrations and the stapes pushes the oval window in and out.
- This action is passed on to the organ of corti, the receptor of hearing, that contains tiny hair cells that translate the vibrations into an electrical impulse that are transmitted to the brain by sensory nerves.

Main Components of the Hearing

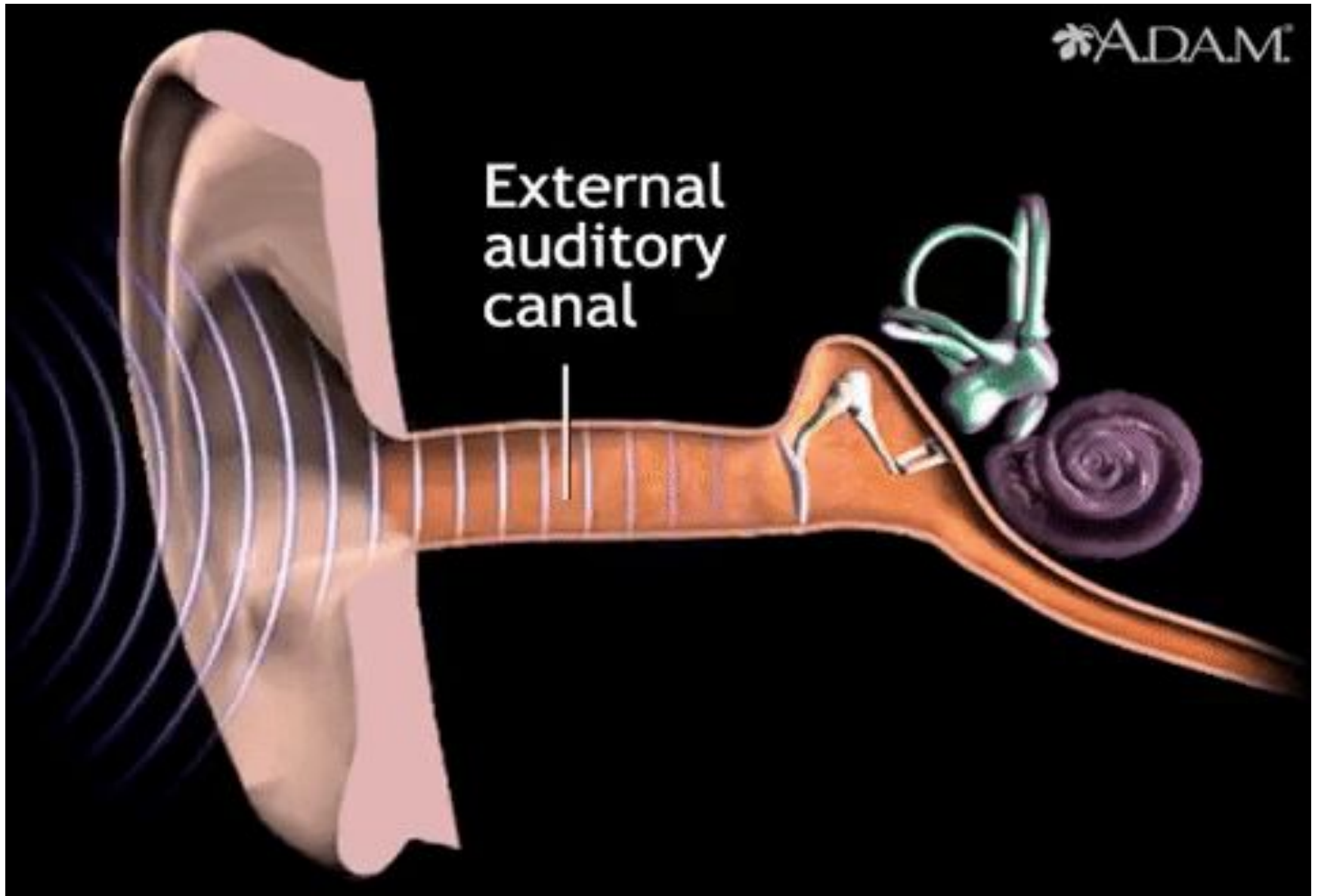
Peripheral

- Outer ear
- Middle ear
- Inner ear
- Auditory nerve

Central

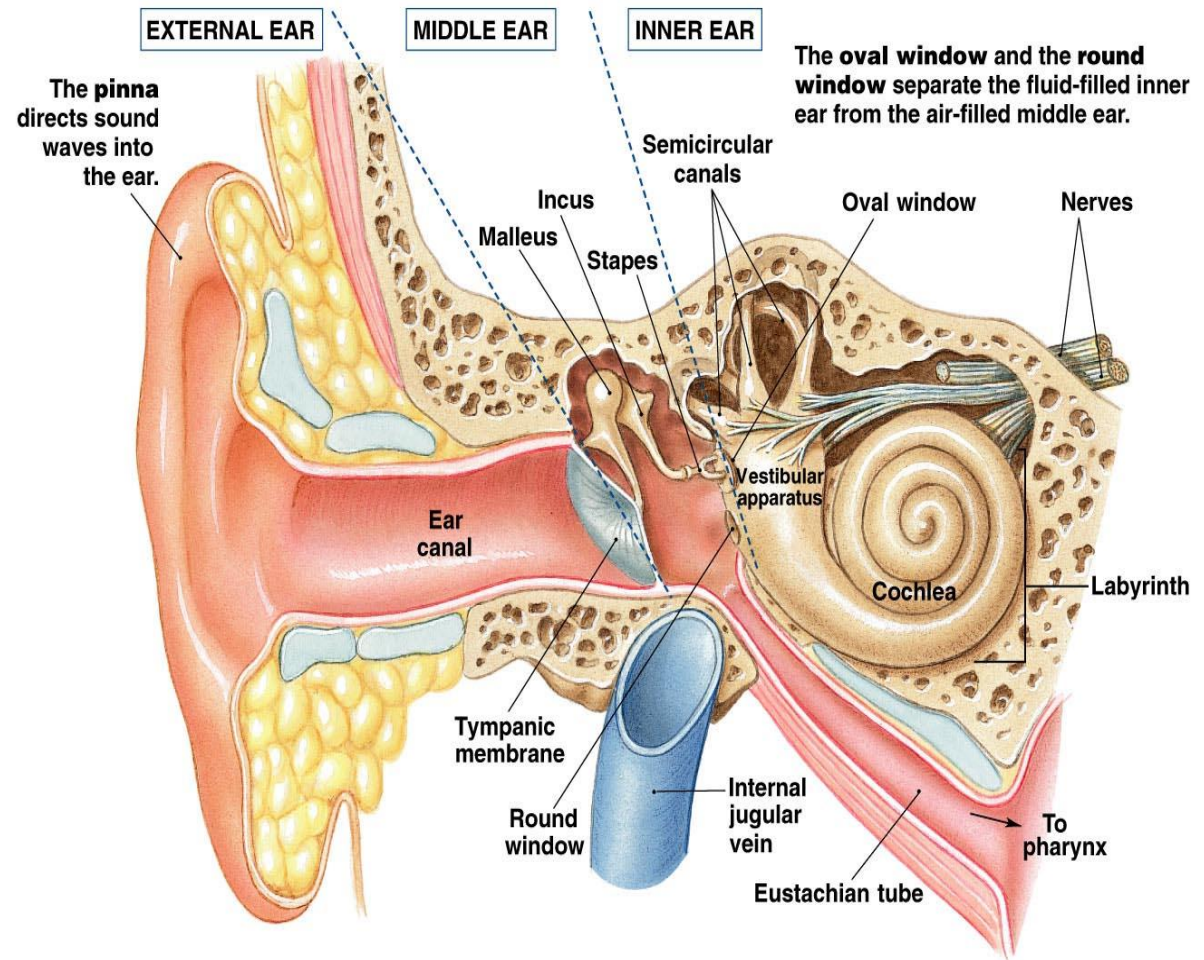
- Brainstem (medulla oblongata)
- Midbrain
- Cerebral cortex

External
auditory
canal



Main Sections

- External Ear
- Middle Ear
- Inner Ear



External Ear

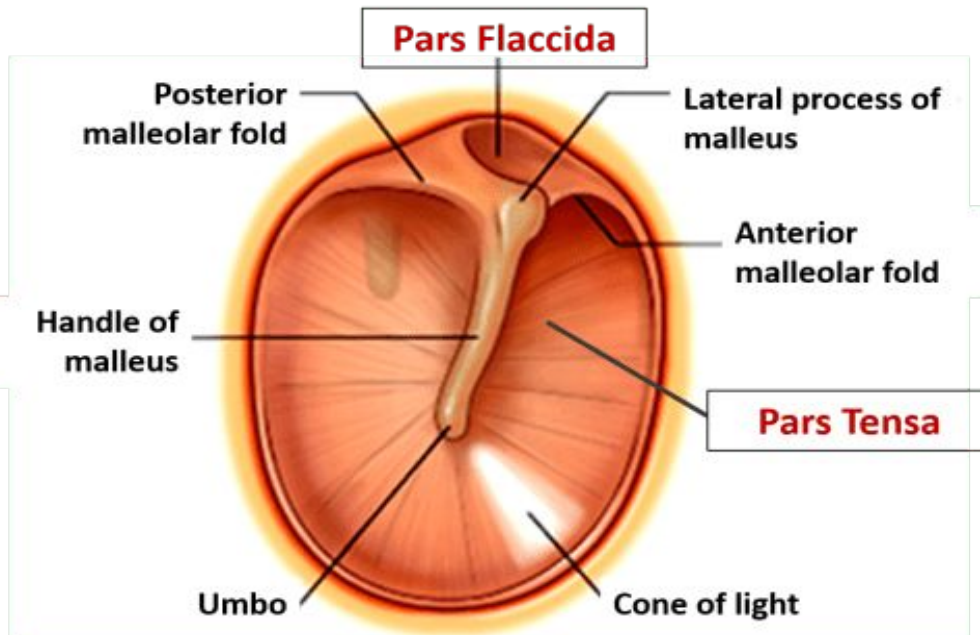
The external ear is further divided into the following parts:

□ **Auricle (Pinna)**

□ **External Auditory Meatus**

□ **Tympanic Membrane**

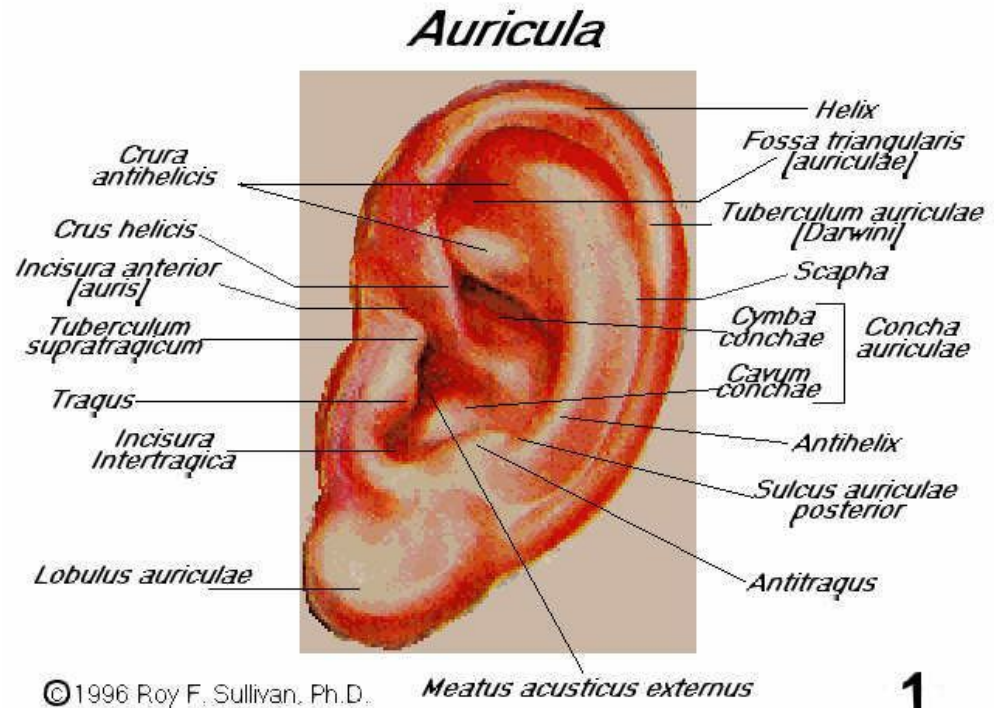
- This membrane separates the middle ear and the external ear.
- This part receives and amplifies the sound waves.
- Its central part is known as the umbo.



Structures of the Outer Ear

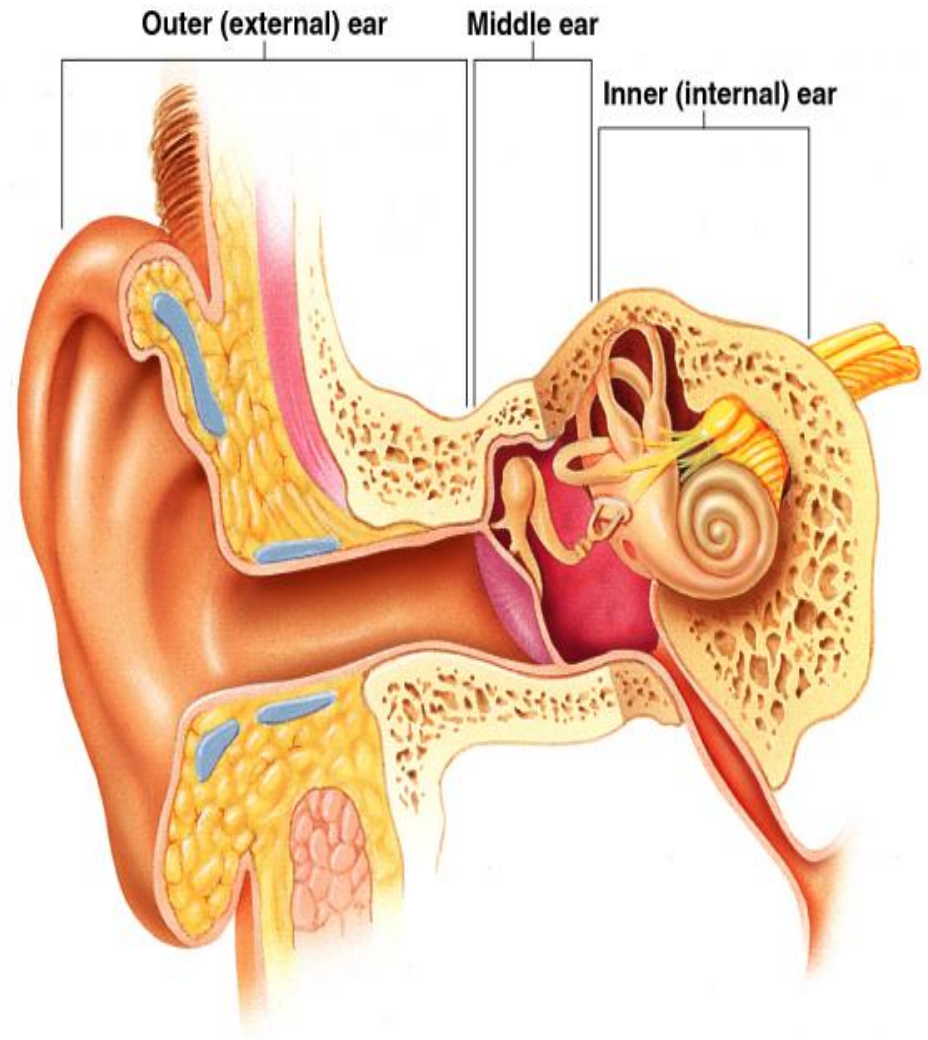
Auricle (Pinna)

- Collects sound and channels them through the air canal.
- Helps in sound localization

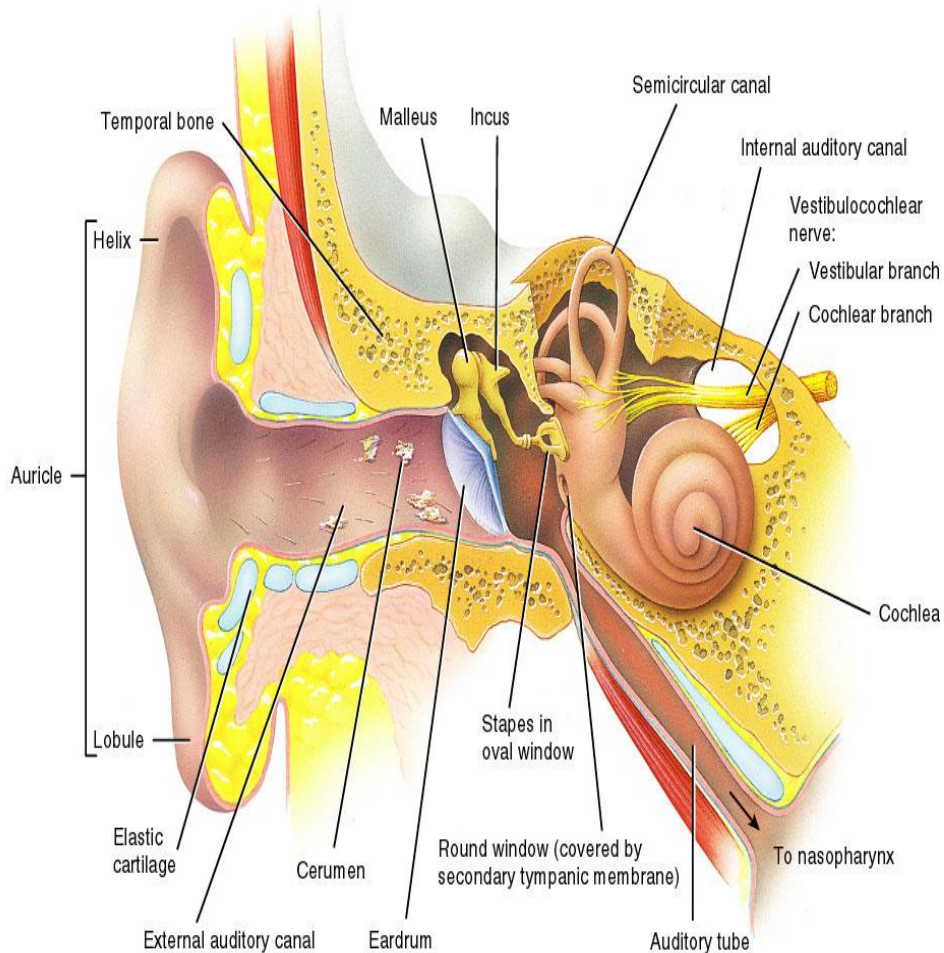


The External Auditory Canal

- Narrow chamber in the temporal bone
- Lined with skin
- Ceruminous (wax) glands are present
- Ends at the tympanic membrane
(eardrum)



Tympanic Membrane



Frontal section through the right side of the skull showing the three principal regions of the ear

16.17

- Thin membrane
- Forms boundary between external and middle ear
- Vibrates in response to sound (moves back & forth)
- Vibrations of tympanic membrane depends upon intensity & frequency of sound waves; vibrating slowly at low frequency, and rapidly in response to high frequency
- Changes sound energy into mechanical energy

Middle Ear

The middle ear comprises the following parts:

Tympanic Cavity

It is a narrow air-filled cavity separated from the external ear by tympanic membrane and from inner ear by the bony wall.

Eustachian Tube

The Eustachian tube is a 4cm long tube that equalizes air pressure on either side of the tympanic membrane.

It connects the tympanic cavity with the nasophary

Ear Ossicles

The middle ear consists of the tympanic membrane and the bony ossicles called the malleus, incus, and stapes.

These three ossicles connect the tympanic membrane to the inner ear allowing for the transmission of sound waves

These are responsible for transmitting sound waves from the eardrum to the middle ear.

Malleus:

A hammer-shaped part that is attached to the tympanic membrane through the handle and incus through the head. It is the largest ear ossicle.

Incus:

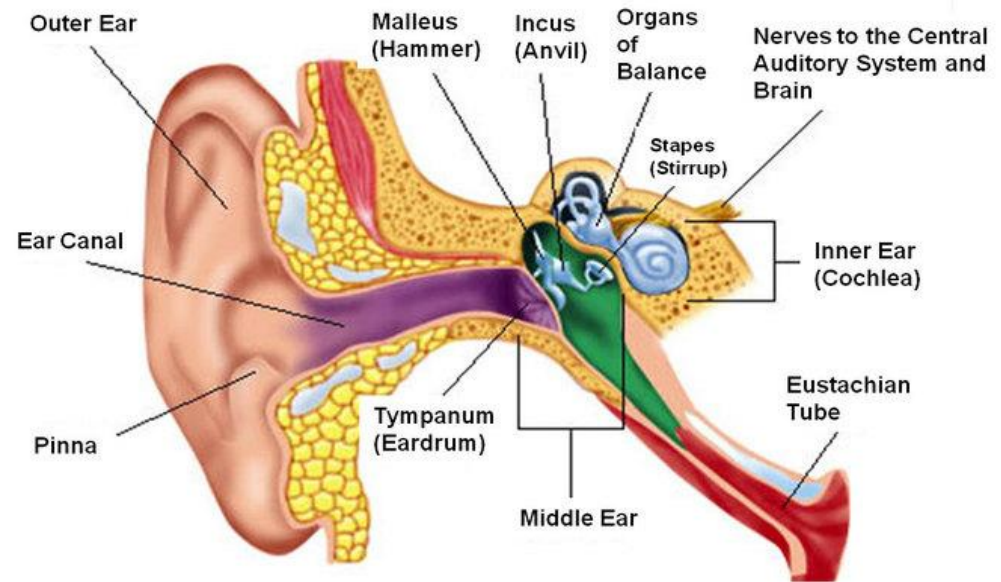
An anvil-shaped ear ossicle connected with the stapes.

Stapes:

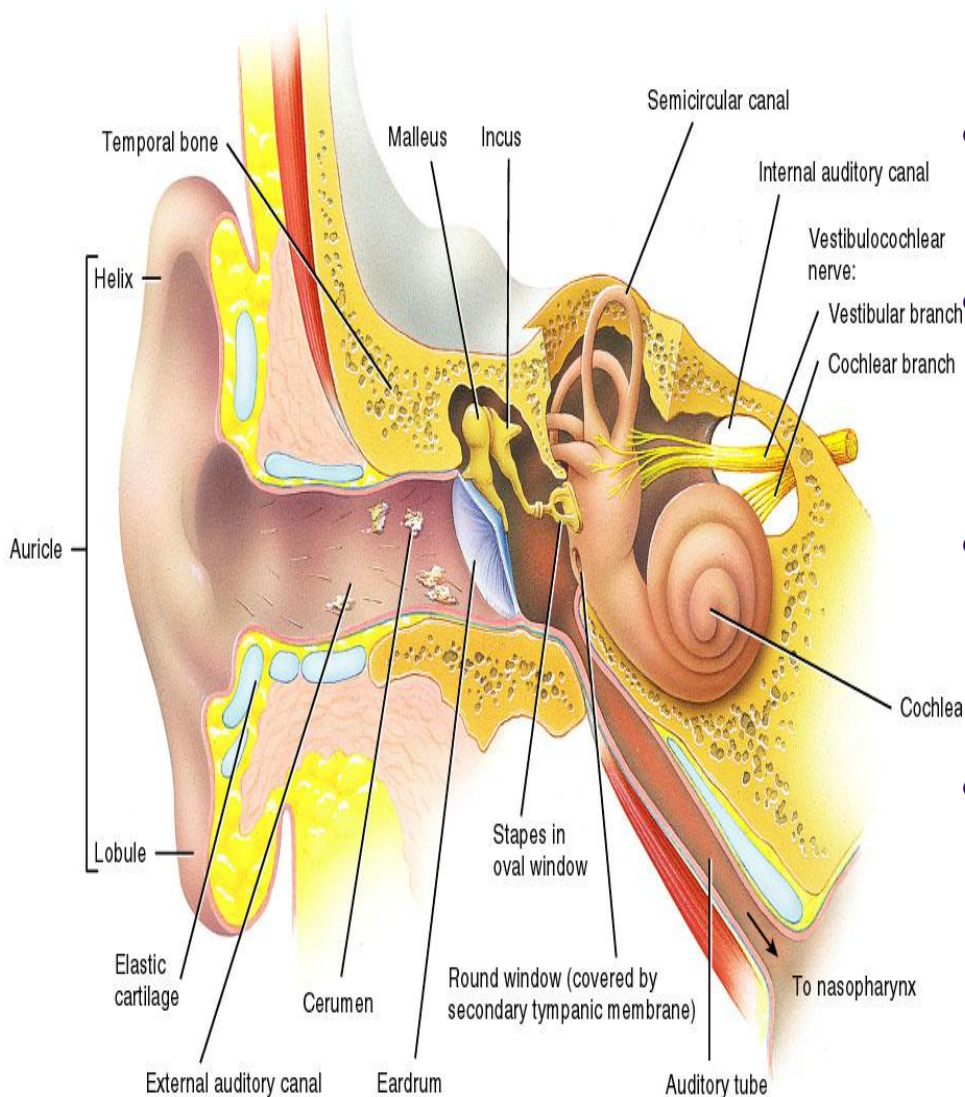
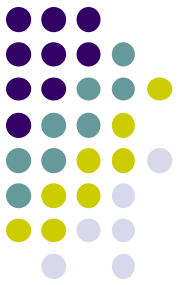
It is the smallest ossicle and also the smallest bone in the human body.

Muscles

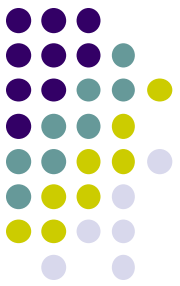
- 1- Tensor tympani
- 2- Stepedius



Eustachian Tube



- Connects middle ear to back of the throat (nasopharynx)
- Equalizes air pressure b/w middle ear and external environment
- Normally closed except during yawning, chewing or swallowing
- when fail to open –ve pressure develops so leads to stiffness in the middle ear mechanical transmission system



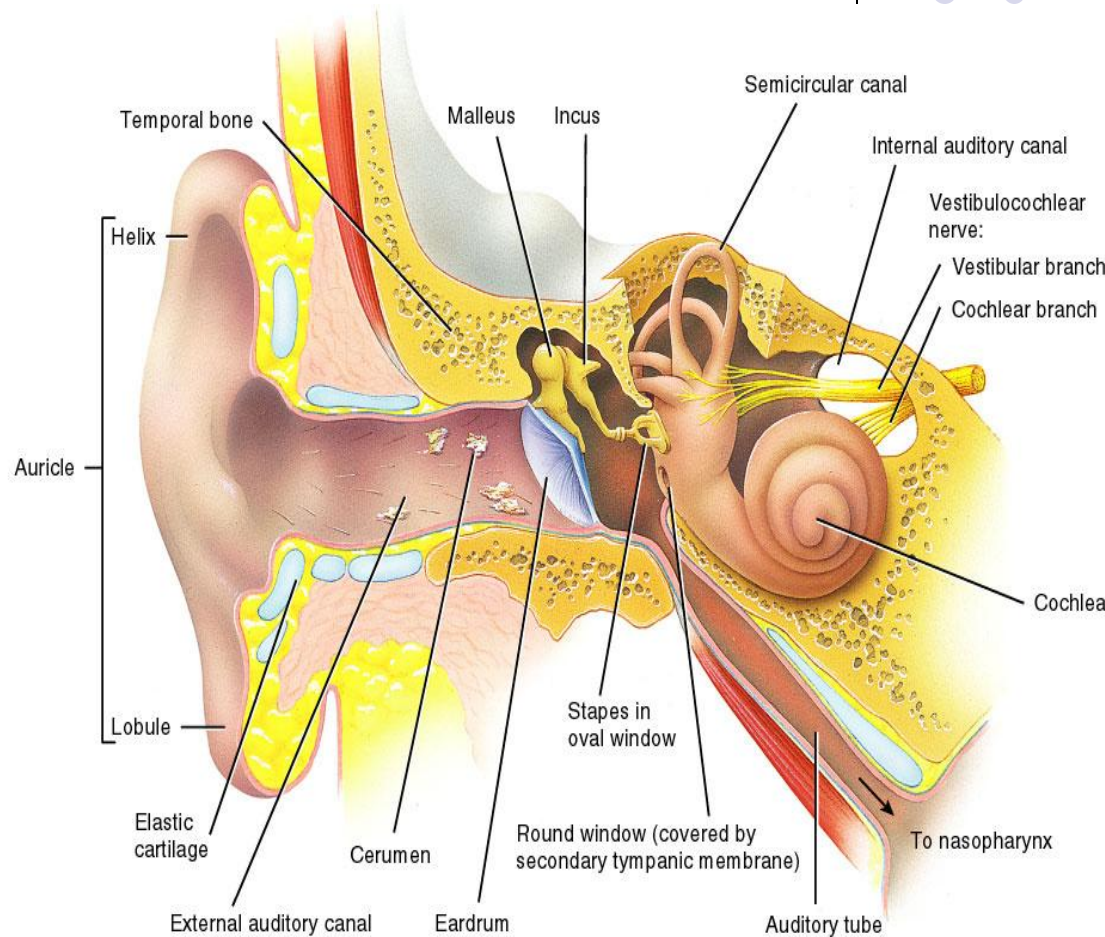
The Ossicular Chain

A: Malleus (hammer)

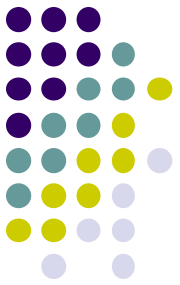
B: Incus (anvil)

C: Stapes (stirrup)

- Ossicles are smallest bones in the body
- Act as a lever system, vibrate in response to sound
- Footplate of stapes enters oval window of the cochlea



Frontal section through the right side of the skull showing the three principal regions of the ear



Functions of middle ear

1. Transmission of sound from external ear to fluid filled inner ear thru ossicular chain while doing it the middle ear performs **impedance matching**
2. Attenuation of sound by **attenuation reflex.**

Inner Ear

It comprises two parts:

- 1-Bony labyrinth
- 2-Membranous labyrinth
- oval window & round window

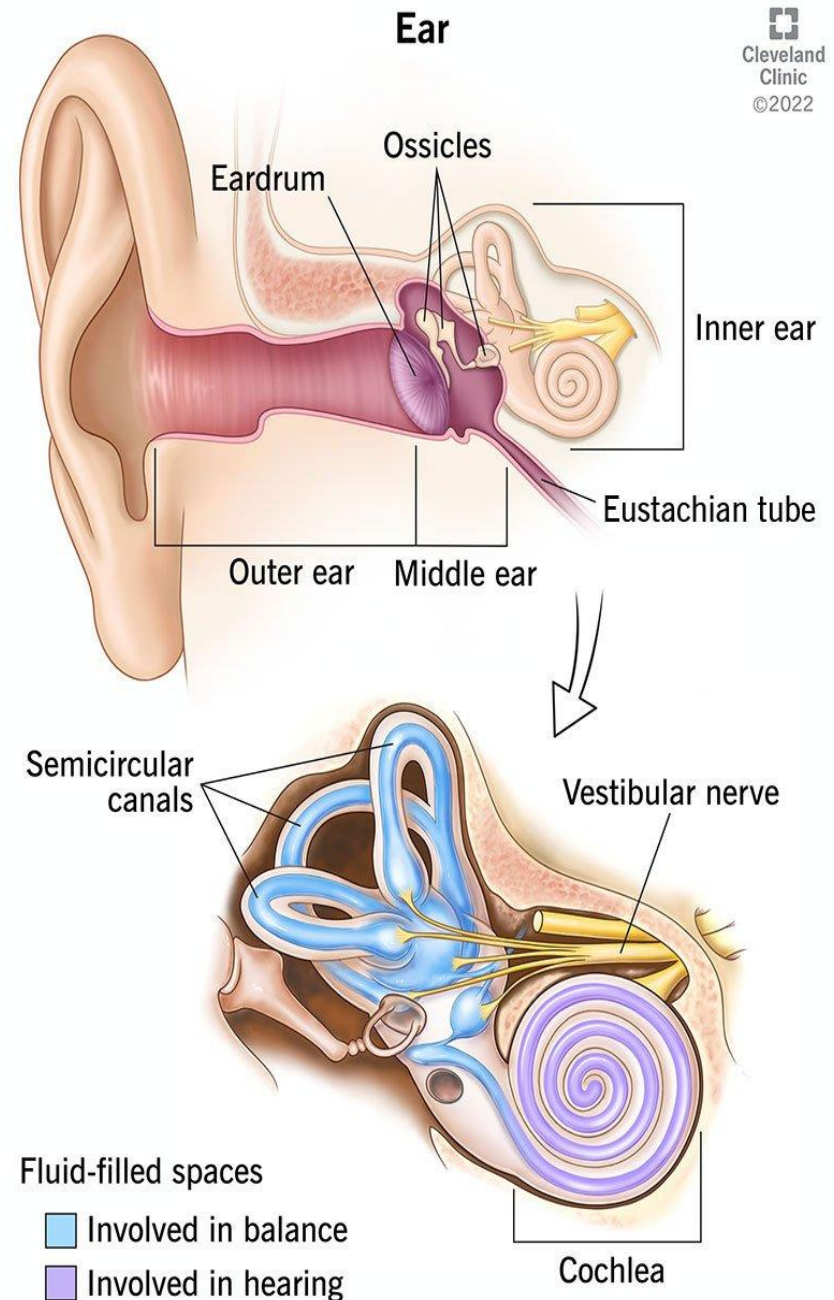
Bony Labyrinth

The bony labyrinth comprises of

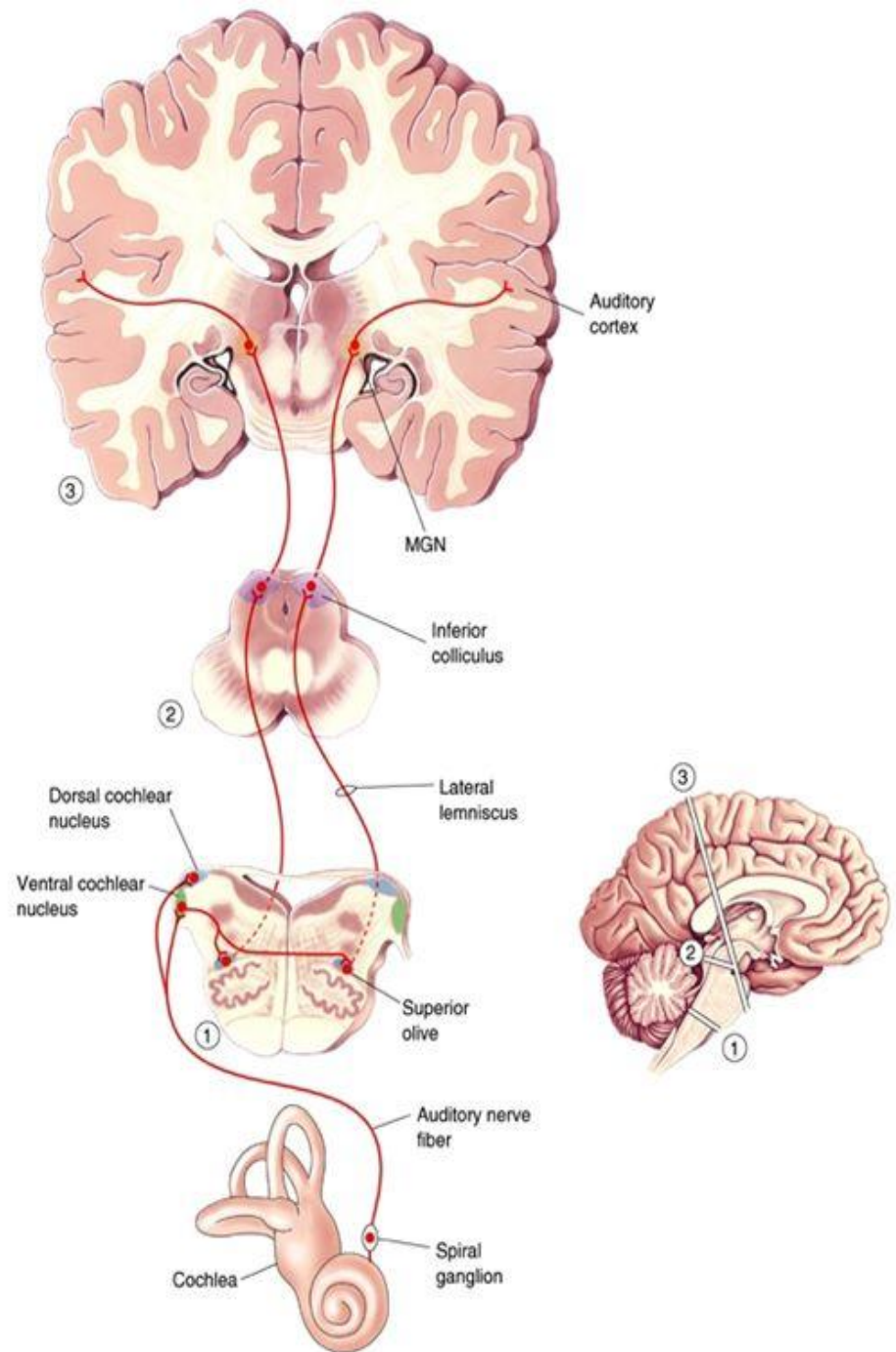
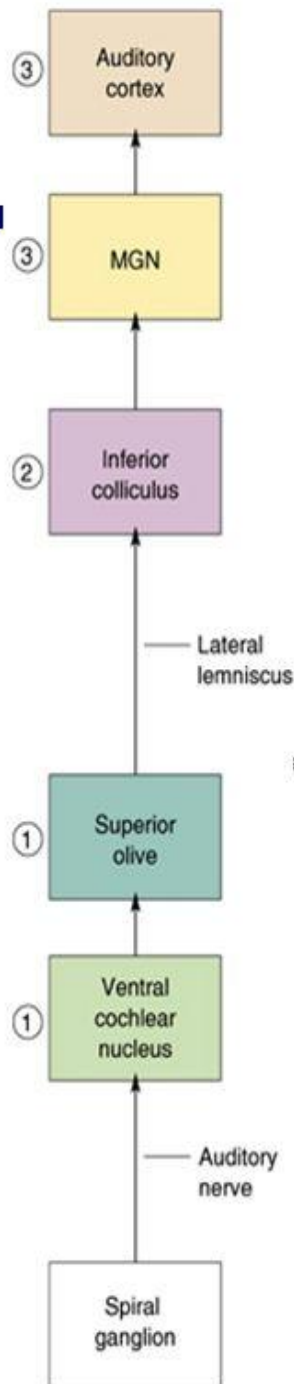
- 1-vestibule,
- 2-three semi-circular canals,
- 3- Cochlea. Spirally coiled. It is filled with perilymph.

Membranous labyrinth

- The bony labyrinth surrounds the membranous labyrinth. It comprises sensory receptors responsible for balance and hearing.



Auditory Pathway



The Ascending Auditory Pathway

Primary Auditory Cortex

Medial Geniculate Body

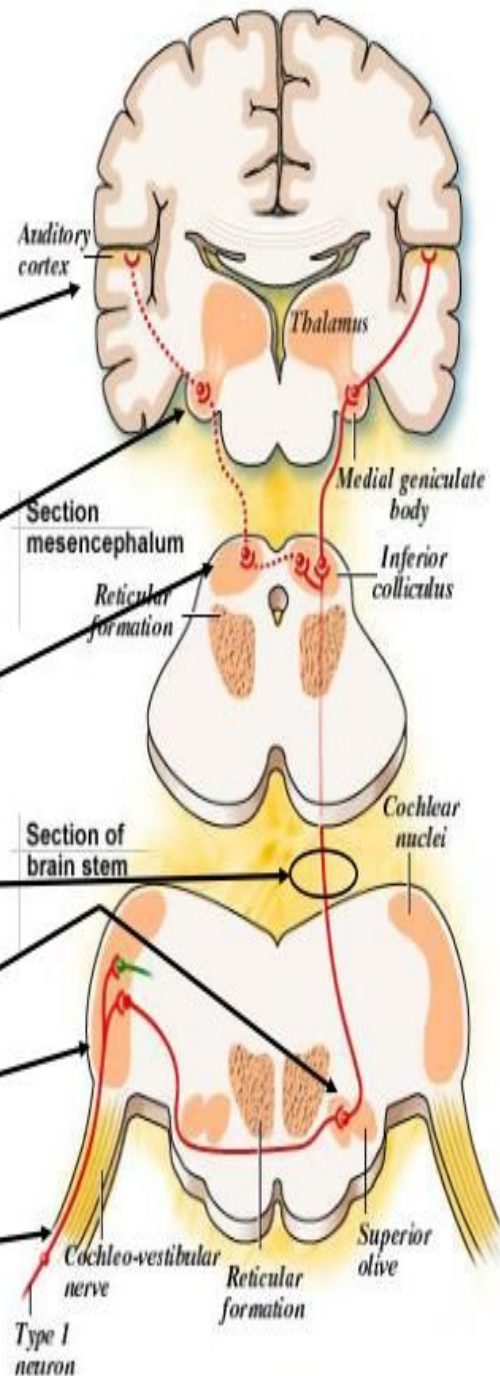
Inferior Colliculus

Lateral Lemniscus

Superior Olivary Complex

Cochlear Nucleus

VIIIth Nerve



HEARING LOSS

Deafness can be

- Conductive or
- Sensor neural



Conductive Hearing Loss

Conductive hearing loss
occurs here in the outer ear.

Inner ear.



CONDUCTIVE HEARING IMPAIRMENT

- A conductive loss refers to a decrease in sound caused by a problem in the outer or middle ear



Conductive deafness

Impairment of sound transmission through external or middle ear due to:

- Wax
- Repeated infection
- Perforated drum
- Destruction of ossicles
- Osteosclerosis (pathological fixation of stapes on the oval window)

All sound frequencies are equally affected

Bone conduction is better than air conduction

Perceptive deafness

Due to congenital or damage to cochlea or auditory nerve pathway due to:

- Toxins (antibiotics, gentamycine)
- Inflammation
- Vascular
- Tumour

Both air and bone conduction are affected

Sensorineural Hearing Loss



SENSORINEURAL HEARING IMPAIRMENT

- A sensor neural loss refers to a problem located in inner ear or along the nerve pathway between the inner ear and the brain



MIXED HEARING IMPAIRMENT

- A mixed loss refers to a conductive loss and a sensor neural loss occurring at the same time



Fig. The Weber test

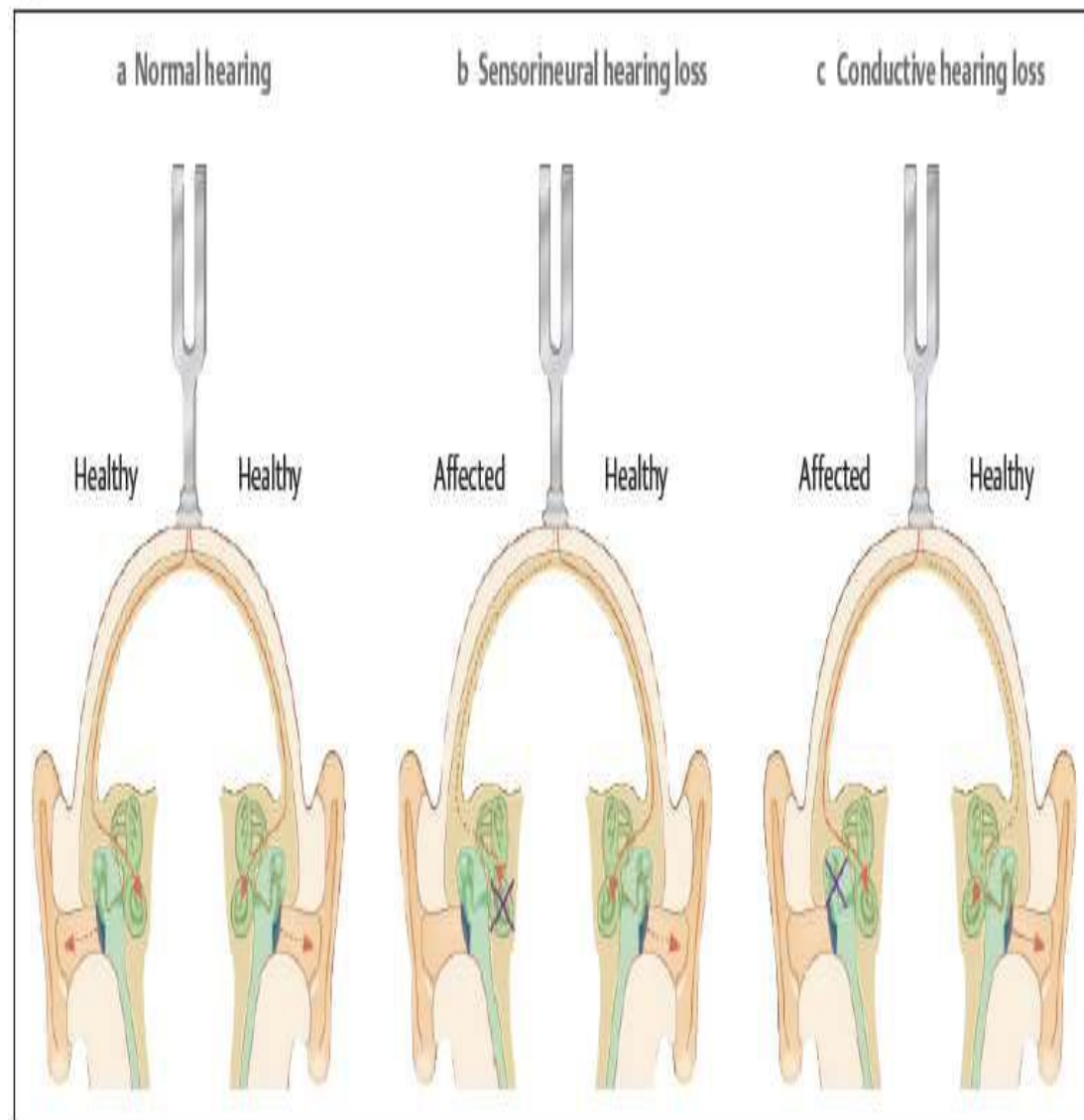
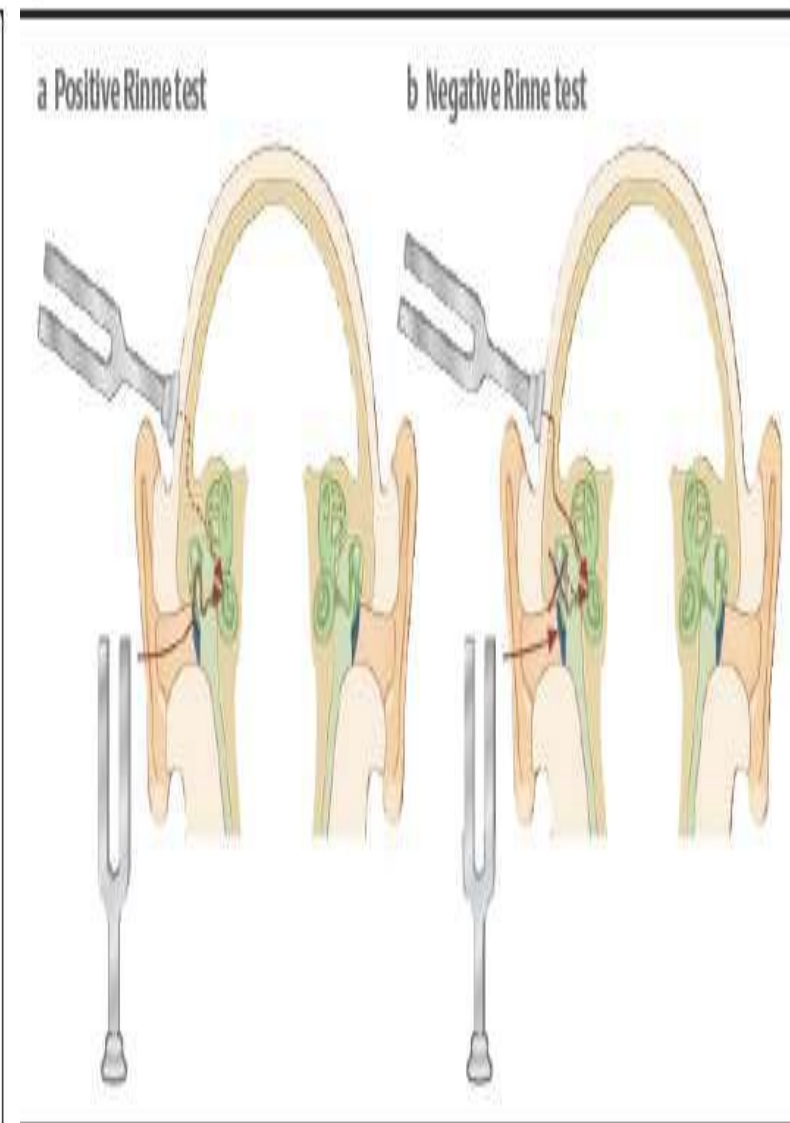


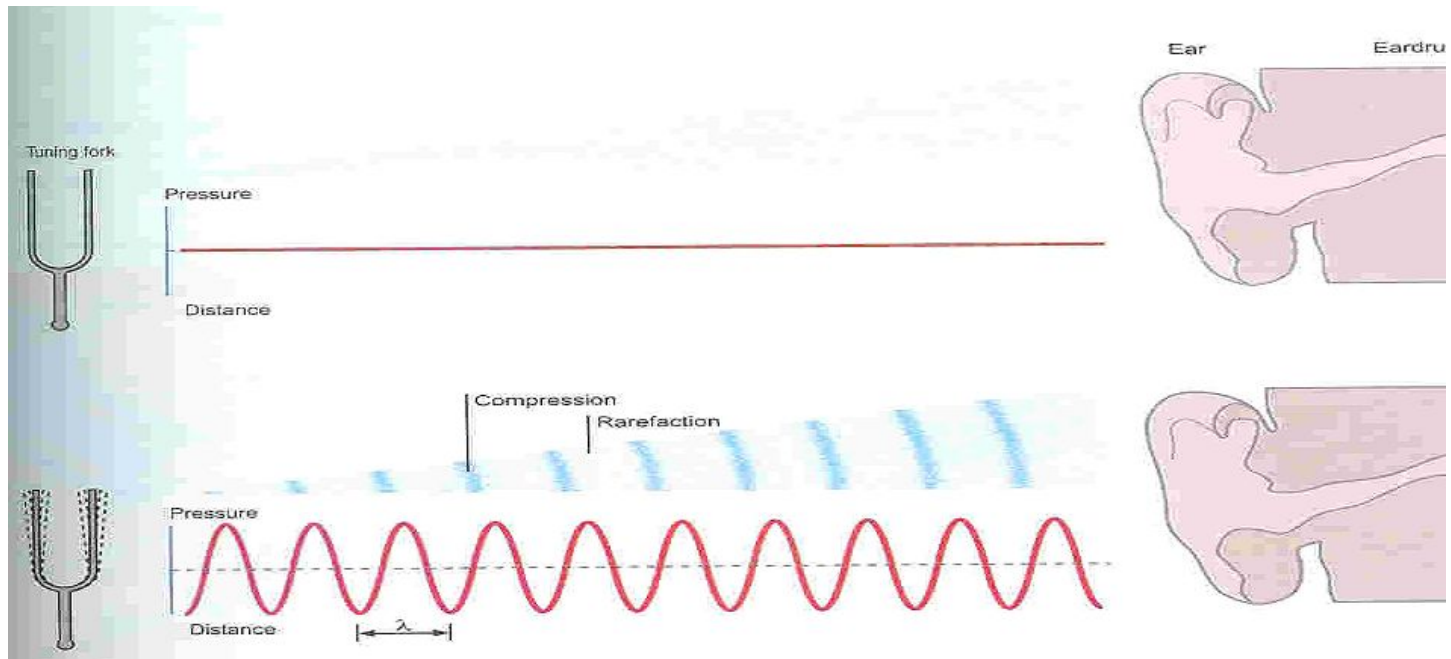
Fig. The Rinne test



Nature of Sound

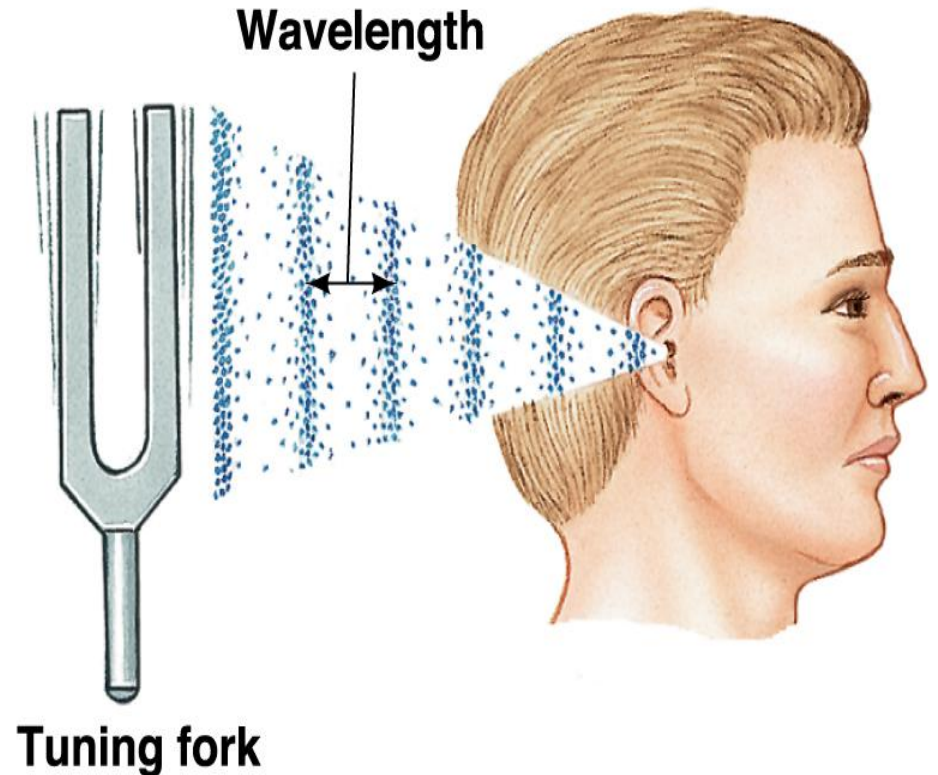
Sound is produced from alternate compression and rarefaction of air molecules by vibrating body's

The total number of waves produced in one second is called the **frequency of the wave**.



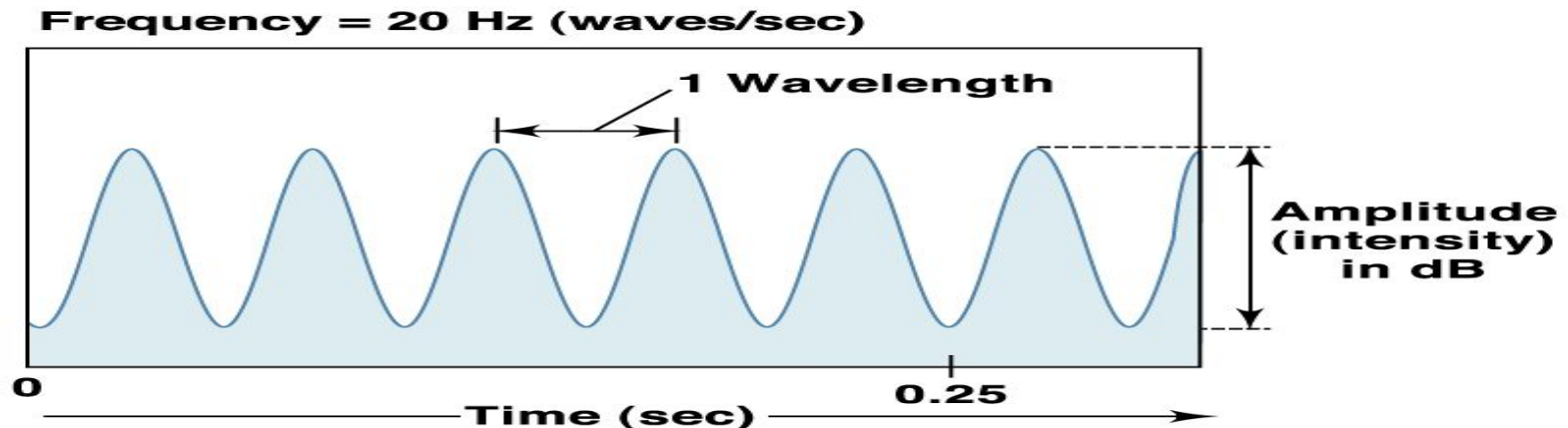
Sound

- Sound waves consist of alternate regions of compression & decompressions of air molecules
- Speed: 340 m/s through air
- Sounds cannot travel through a vacuum.



Properties of Sound

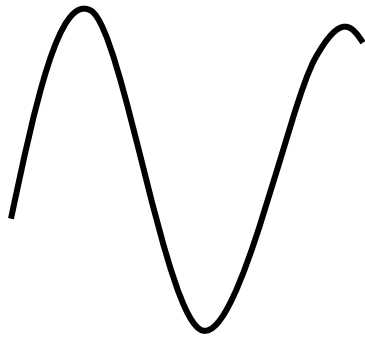
- **Pitch** = Frequency of waves arriving measured in hertz (Hz)
- The frequencies of sound a young person can hear are between 20 and 20,000 cycles/sec
- **Loudness** = Intensity of waves measured in decibels (dB)
Intensity (Loudness) depend on amplitude



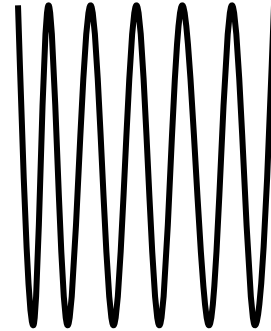
Physics of Sound

Frequency –

Number of cycles completed by a wave in a given amount of time



Low Frequency

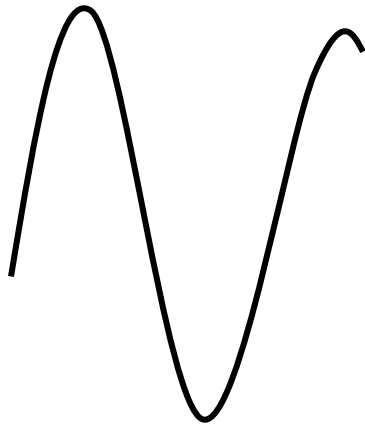


High Frequency

Physics of Sound

Amplitude –

Physical strength of a wave



High Amplitude



Low Amplitude



Pitch, Loudness, and Timbre

Pitch –

Sensory characteristic of sound produced by the frequency of the sound wave

Loudness –

Sensory characteristic of sound produced by the amplitude (intensity) of the sound wave

Timbre –

Quality of a sound wave that derives from the wave's complexity

Pitch (tone)

depends on
frequency



Low note



High note

Intensity (loudness)

depends on amplitude



Soft



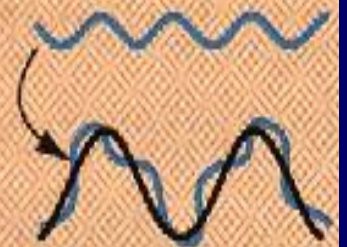
Loud

Timbre (quality)

depends on
overtones



Pure tone



Different overtones

PRACTICAL: TO PERFORM THE **HEARING ACUITY TESTS**

OBJECTIVES

By the end of the session, the students should be able to:

- Differentiate between the types of deafness
- Perform the hearing tests
- Interpret the results

REQUIREMENTS

- Sound proof room
- Tuning fork
- Human subject
- Cotton wool

Signs of hearing loss



Misunderstanding
people

Difficulties on
the telephone

Turning the TV
or radio up

Ringings in the
ear or head

Cannot hear high
pitched sounds

Speaking
loudly

Asking people to
repeat themselves

Problems in noisy
environments

TUNING FORK

- A small metal instrument consisting of stem, two prongs and a foot piece that produces a constant pitch when struck against a firm but resilient surface
- It is usually made of steel, magnesium or aluminium that is used to tune musical instruments by musicians and now used by doctors to assess hearing
- It vibrates sinusoidally to generate pure tone



TUNING FORK TEST

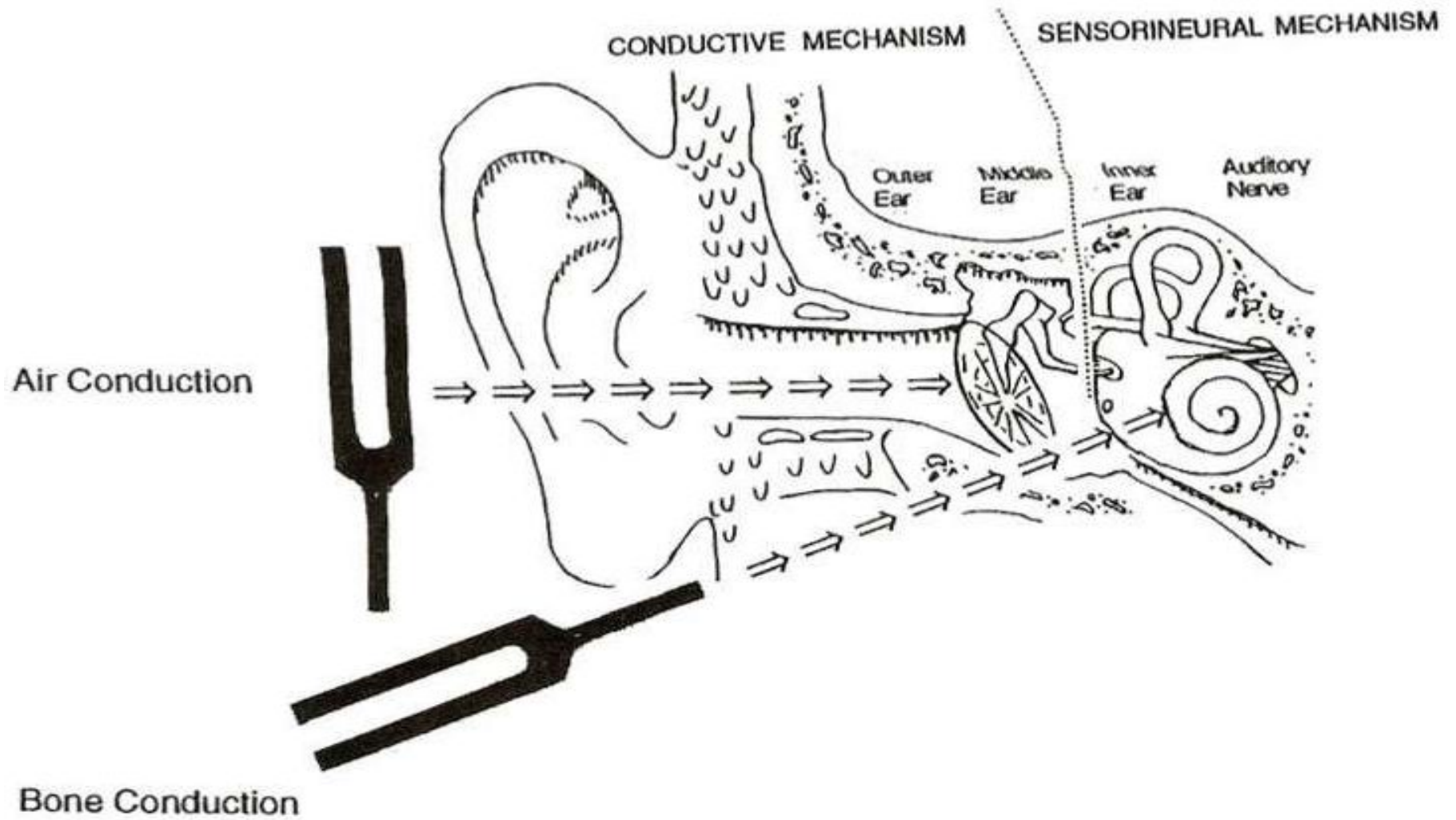
- Used to differentiate between conductive and sensorineural hearing loss
- Test is performed with different types of frequency like **128, 256, 512, 1024, 2048** Hz etc. In routine practices, 256, 512 and 1024 Hz forks are used
- Larger tuning forks vibrate at slower frequency
- Tuning fork is activated by striking against the examiner's elbow, heel of the hand or against a rubber pad and placed 2cm away from external auditory canal for Air conduction and on mastoid process for Bone conduction

AIR CONDUCTION (AC)

- Vibrating tuning fork is placed vertically in line with the meatus about 2cm away from the external auditory canal opening
- The sound waves transmitted through Tympanic Membrane- Middle ear ossicles – Auditory Nerve – Auditory cortex
- Both conductive mechanism and cochlea are tested

BONE CONDUCTION (BC)

- Foot plate of vibrating tuning fork is placed on the mastoid process
- Cochlea is stimulated directly by the vibrations conducted through the skull
- Bone Conduction measures only the cochlear functions
- Normally AC will be more than BC



PRINCIPLES OF TUNING FORK TESTS

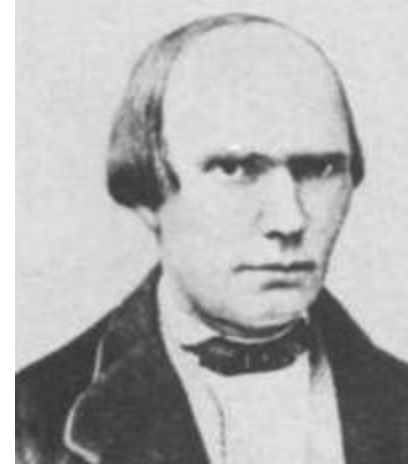
- Conducting Hearing Loss (CHL)
 - Sounds delivered to the ear via AC will be decreased
 - If sound is delivered to the ear via BC, bypassing the outer ear and middle ear, then the sound will be heard normally, assuming there is no disorder
- Sensorineural Hearing Loss (SNHL)
 - Sounds delivered to the ear will be decreased

TYPES

- Rinnie's
- Weber's
- Schwabach's
- Modified Schwabach (A.B.C.)

RINNE'S TEST

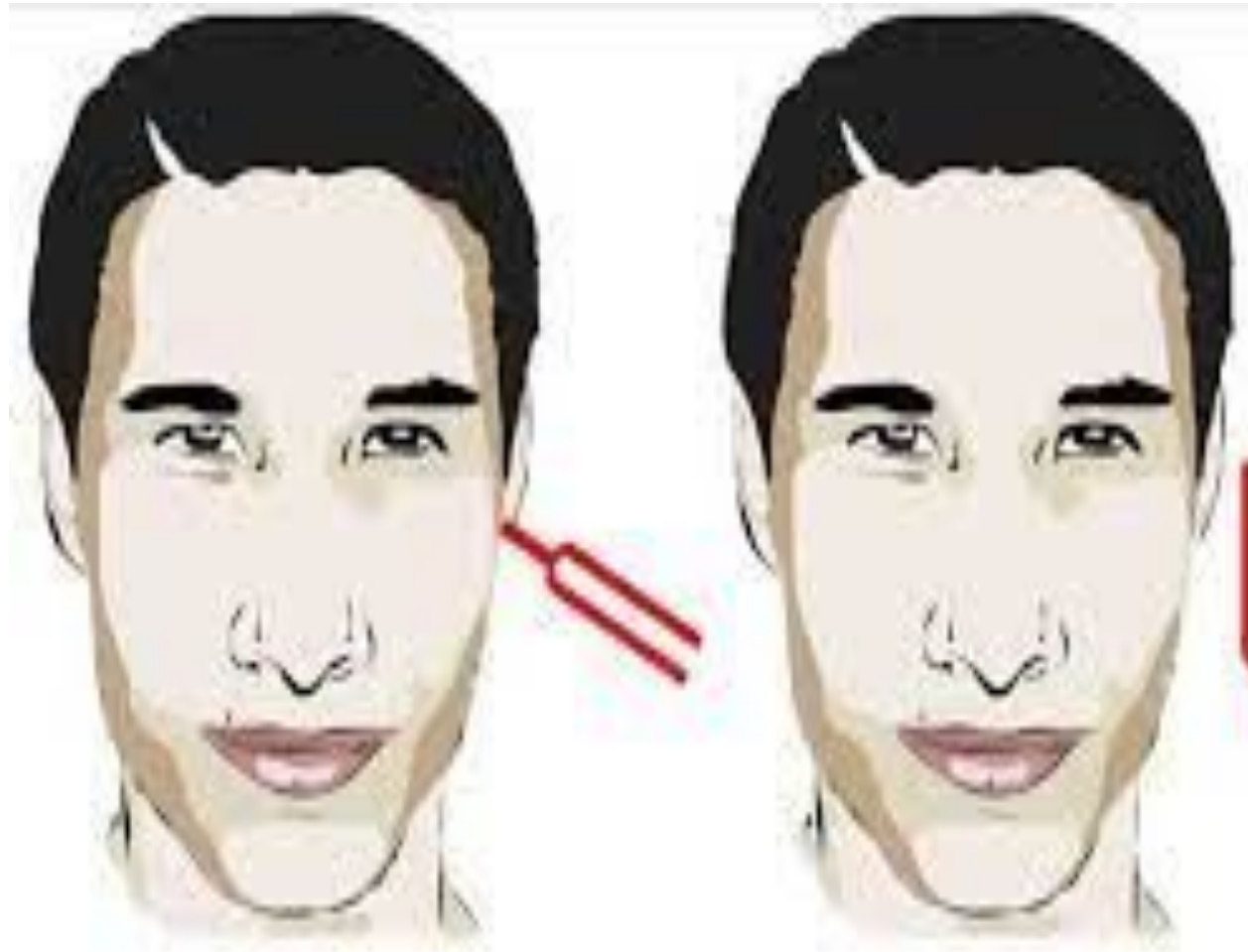
- **PRINCIPLE: COMPARES THE AC WITH BC OF THE SAME EAR. AC IS BETTER THAN BC 3:1**
- **Proposed by Heinrich Adolf Rinne (1855)**
- Air conduction is compared with bone conduction
- The base of the vibrating fork is placed firmly over the mastoid process
- Patient is asked to indicate when sound disappears. Suddenly the tuning fork is placed vertically 2cm from External auditory canal
- If still hears: AC more than BC
- Rinne's positive i.e. Normal



PROCEDURE

- Strike tuning fork with rubber pad
- Place it in front of the External Auditory Meatus
- Subject should raise the index finger once he stops hearing the sound
- Put it immediately on the mastoid process and observe if the subject can hear any sound. If able to hear, Rinnie's is negative
- Reverse the process. Put the tuning fork first on the mastoid process. Once subject stops hearing it, immediately transfer it to the front of mastoid process
- The subject should hear the sound

RINNE'S TEST



Rinne's Test

With a 512 Hz tuning fork press against the mastoid bone and then hold it 1cm away from the ear.

'Which is louder, behind the ear or in front?'

INFERENCE

- Rinne's Positive
 - Air Conduction > Bone Conduction = Normal or SNHL
- Rinne's Negative
 - Bone Conduction > Air Conduction = Conductive Hearing Loss
- False Negative Rinne's
 - Air Conduction absent, but responds to Bone Conduction
 - Response of bone conduction is due to transcranial transmission from the opposite ear
 - Seen in UNILATERAL SENSORINEURAL HEARING LOSS

WEBER TEST

- Proposed by Ernst Heinrich Weber
- A test of lateralization
- A vibrating tuning fork is placed in the middle of the forehead or the vertex, then asked in which ear the sound is heard
- Normally, heard equally in both ears
- Lateralized to the worst ear in conductive hearing loss
- Sound travels directly to the cochlea through the bone



WEBER TEST

- Based on “**Stenger’s Principle**”:
- If two identical tones, except their difference in loudness, are introduced simultaneously in both ears; only the louder tone will be perceived



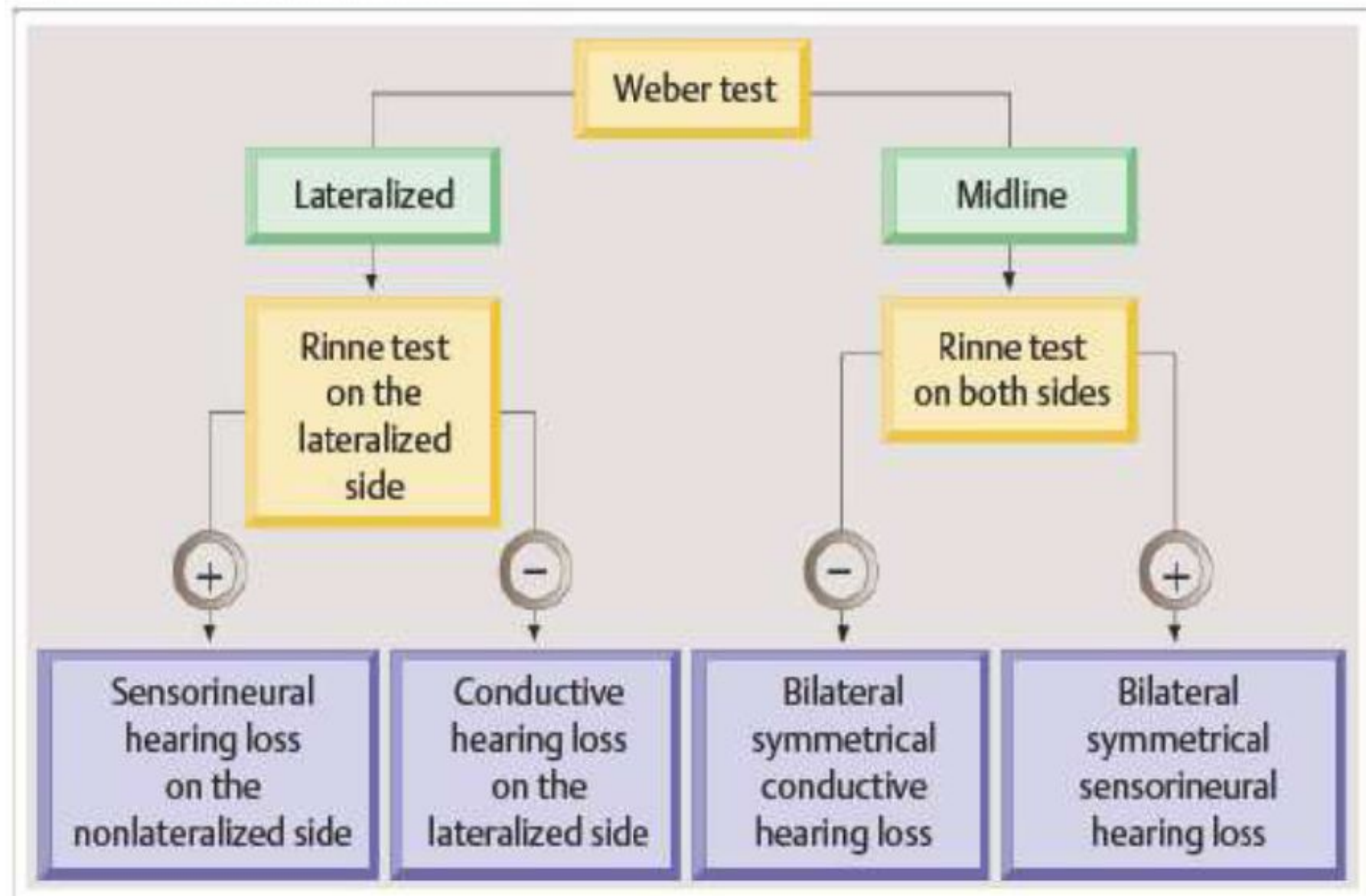
WEBER'S TEST: PROCEDURE

- **PRINCIPLE:** Compares the Bone conduction of one ear with the other ear
- Strike the tuning fork on the rubber pad and place it on the vertex
- Ask the subject if he is able to hear it equally on both sides

INFERENCE

- If the tone lateralizes to the poorer ear:
 - Conductive deafness in the poorer ear
- If the tone lateralizes to better / good ear:
 - The cochlea with the best hearing sensitivity will detect the signal
 - SNHL in the poorer ear
- If the sound is detected in the midline position:
 - Normal hearing or
 - Equal amounts of same type of hearing loss in both the ears (Conductive or Sensorineural)

Fig. Tuning fork tests



SCHWABACH'S TEST

- Once popular but no longer in use
- Compares the patient's hearing sensitivity with that of an examiner
- The fork is set into vibration; stem is placed alternately against the mastoid process of the patient and the examiner. Here, meatus is not closed
- Vibratory energy of the tines of the fork decrease overtime, making the tone softer
- Patient should indicate whether the tone is heard or not each time
- When the patient no longer hears the tone, the examiner immediately places the stem behind his or her own ear and using a watch, notes the number of seconds the tone is audible after the number of seconds the patient stops hearing it

SCHWABACH'S TEST: PROCEDURE

- **PRINCIPLE:** Compares the bone conduction of the subject with the examiner
- Strike the tuning fork with the rubber pad
- Put it on the subject's mastoid process. Ask him to raise index finger when sound becomes inaudible
- Immediately transfer the tuning fork to the examiner's mastoid process
- Observe if the examiner can hear or not
- Record the time in seconds

INFERENCE

- Normal Schwabach
 - Both patient and examiner stop hearing the tone at approximately the same time
 - Patient has normal BC
- Conductive Deafness
 - If time in case of the subject is lengthened than the examiner
- Sensorineural Hearing Loss
 - If time in case of the subject is lengthened than the examiner
 - Patient stops hearing the sound sooner than the examiner

Modified Schwabach's Test / Absolute Bone Conduction (A.B.C. Test)

- Modification of Schwabach Test
- Patients' bone conduction is compared with that of the examiner (assuming that he / she has a normal hearing)
- External auditory canal is occluded by pressing the tragus inwards, to prevent the ambient noise entering through AC route

ABC TEST: PROCEDURE

- Strike the tuning fork with the rubber pad
- Put it on the subject's mastoid process with the meatus occluded, and ask him to raise index finger when sound becomes inaudible
- Immediately transfer the tuning fork to the examiner's mastoid process
- Observe if the examiner can hear or not
- Record the time in seconds

INFERENCE

- Normal: Bone conduction of patient is equal to the examiner
- In SNHL, ABC is reduced compared to the examiner

Table 9–1. Common Tests with a Tuning Fork to Distinguish between Nerve and Conduction Deafness.

	Weber	Rinne	Schwabach
Method	Base of vibrating tuning fork placed on vertex of skull.	Base of vibrating tuning fork placed on mastoid process until subject no longer hears it, then held in air next to ear.	Bone conduction of patient compared with that of normal subject.
Normal	Hears equally on both sides.	Hears vibration in air after bone conduction is over.	
Conduction deafness (one ear)	Sound louder in diseased ear because masking effect of environmental noise is absent on diseased side.	Vibrations in air not heard after bone conduction is over.	Bone conduction better than normal (conduction defect excludes masking noise).
Nerve deafness (one ear)	Sound louder in normal ear.	Vibration heard in air after bone conduction is over, as long as nerve deafness is partial.	Bone conduction worse than normal.

Table 20.1 Deafness

Conductive	Sensorineural
Congenital – atresia Pendred's syndrome (see p. 1071) Long QT syndrome (see p. 777)	End organ Advancing age Occupational acoustic trauma Ménière's disease Drugs (e.g. gentamicin, furosemide (frusemide))
External meatus Wax Foreign body Otitis externa Chronic suppuration	Eighth nerve lesions Acoustic neuroma Cranial trauma Inflammatory lesions: Tuberculous meningitis Sarcoidosis Neurosyphilis Carcinomatous meningitis
Drum Perforation/trauma	
Middle ear Otosclerosis Ossicular bone problems Suppuration	Brainstem lesions (rare) Multiple sclerosis Infarction

Relevant Questions

Q-1-What is the role of Ossicles in process of hearing?

The ossicles (malleus, incus, and stapes) are three tiny bones in the middle ear that act as a bridge between the eardrum and the inner ear, amplifying and transmitting sound vibrations from the eardrum to the fluid-filled cochlea. This mechanical process converts sound waves from the air into a form that the fluid inside the inner ear can respond to, allowing the brain to process the sound.

Q-2-How is sound wave conducted from the tympanic membrane to the auditory cortex?

From the tympanic membrane, sound waves are conducted through the ossicles (malleus, incus and stapes) in the middle ear, which amplify and transmit the vibrations to the fluid-filled cochlea in the inner ear.

Inside the cochlea, these vibrations are converted into electrical signals by hair cells, which then travel along the auditory nerve to the brainstem and thalamus before reaching the primary auditory cortex in the temporal lobe for interpretation.

Q-3-What do you mean by the terms, Conductive Deafness and Nerve Deafness?

Conductive deafness (also called transmission deafness) indicates that there is an obstacle to sound transmission between the middle ear and the cochlea. Sensorineural deafness (perception deafness) signals a deterioration of the auditory pathway from the cochlea to the auditory cortex of the brain.

Q-4-How can you differentiate between the bone and nerve deafness with the help of tuning fork test?

Tuning fork tests have been the mainstay of otologic examination for more than a century. The Weber test has been mainly used to establish a diagnosis in patients with unilateral hearing loss to distinguish between conductive and sensorineural hearing loss.

Q-5-What is masking of sound?

Sound masking is ambient background sound engineered to match the frequency of human speech for greater speech privacy.

Adding sound to a space actually makes the space seem quieter. It sounds counter-intuitive but it's true. This is because the added sound reduces the intelligibility of human speech.